

Non-additive Strain–Dopant Energetics in Quasi-Hexagonal C₆₀ Graphullerene

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Under mechanically constrained loading on quasi-hexagonal (qHP) C₆₀ graphullerene, additive strain–dopant energetic screening fails: totals on the joint $E(\epsilon, \delta)$ landscape need not equal the sum of separate strain-only and substitution-only increments, and the linear strain coefficient α does not reliably predict the omitted bilinear cross term. Using matched PBE+D3 rigid-lattice protocols on B/N/P tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$), matched four-corner audits show that the mixed second-order coupling $c\epsilon\delta$ in Eq. (1) is nonzero and, for size-mismatched phosphorus, large enough to change high-throughput stability rankings ($|\mathcal{S}| = 31.9 \pm 2$ meV/atom at $n=1$, $\eta \approx 8\%$ at $n=4$). The four-corner difference $\mathcal{S}$ is a finite-difference *estimator* of that term [Eqs. (4)–(3)], not a new thermodynamic observable. Non-additivity scales with frozen local elastic incompatibility ($|\Delta r_{\text{cov}}| \gtrsim 20$ pm) rather than dopant identity alone; ionic relaxation nearly extinguishes $|\mathcal{S}|$, indicating that the audited coupling originates predominantly from stored elastic work under constrained load rather than from an intrinsic electronic interaction term. The audit signals when sequential additive workflows should be replaced by explicit coupled evaluations.

I. INTRODUCTION

Substitutional doping and mechanical strain are widely used strategies for tuning energetic stability and electronic structure in covalent molecular networks [1, 2]. While external biaxial strain modifies global bonding geometry, substitutional dopants introduce localized chemical stress, lattice distortion, and bond competition, so the two perturbations can become energetically coupled through higher-order cross terms in the total-energy expansion [3]—i.e., the total response need not separate into strain-only and composition-only increments without a nonlinear interaction term. Configurational interaction and cluster-expansion effective Hamiltonians for alloys already include explicit composition–strain cross terms [3], so a bilinear correction to $E(\epsilon, \delta)$ is expected in principle. Expanding the energy per atom $e(\epsilon, x) = E(\epsilon, \delta)/N_{\text{atoms}}$ about the pristine reference gives, schematically,

$$e(\epsilon, x) \approx e_0 + a\epsilon + bx + c\epsilon x + \mathcal{O}(\epsilon^2, x^2, \epsilon^2 x, \epsilon x^2), \quad (1)$$

where $c\epsilon x$ is the lowest-order non-additive coupling between external strain and composition; at finite load, $c = \partial^2 e / \partial \epsilon \partial x$ is the bilinear coefficient whose relevance we test. The four-corner estimator $\mathcal{S}$ (Sec. II C) samples $\partial^2 E / \partial \epsilon \partial \delta$ at +3% biaxial load—it is a diagnostic, not the physical claim. In many high-throughput mapping studies, strain and substitution are nevertheless evaluated in separate density-functional theory (DFT) workflows and combined additively, in part because exhaustive coupled (ϵ, δ) grids are computationally expensive [1, 4, 5]. Although first-principles work has extensively mapped strain and doping separately across molecular and two-dimensional networks [5–7], matched four-corner quantification of their total-energy coupling under simultaneous load has rarely been reported for fullerene-based covalent molecular networks. We study

quasi-hexagonal (qHP) C₆₀ graphullerene [8, 9] as an experimentally accessible model covalent molecular network. Prior density-functional work largely falls into two classes: additive mapping studies that sample strain and substitution on separate grids [5–7, 9, 10], and reports of nonlinear strain–composition response in band edges, transport, or magnetism without matched four-corner *total-energy* audits [4, 11, 12]. A literature-positioning summary is given in Supplemental Material, Table I. What remains unresolved for soft graphullerene cages is how large the bilinear correction is, when it reverses *relative energetic stability* and dopant-site preference under simultaneous load—quantities that govern processing and defect retention rather than gap or mobility shifts alone—and when sequential additive workflows become unreliable. Ab initio models of pristine qHP networks report intrinsic intramolecular strain [13]. A substitutional defect imposes a local chemical-stress and bond-distortion field on a pre-strained, elastically soft cage; together with global biaxial load, this motivates a periodic-scale audit of dopant–strain coupling beyond linear α .

Here we demonstrate that additive strain–dopant energetics fail under mechanically constrained joint load on qHP C₆₀: the bilinear coefficient in Eq. (1) is nonzero ($\gamma \neq 0$ in the discrete four-corner sense) and, for phosphorus, large enough to invalidate sequential screening decisions at meV-per-atom margins relevant to stability rankings. On the total-energy landscape $E(\epsilon, \delta)$ of substitutionally doped qHP C₆₀ under biaxial load—where *relative energetic stability*, not band-edge shifts alone, governs accessible substitutional configurations—we test three linked hypotheses: (i) additive totals fail under joint (ϵ, δ) load; (ii) the omitted correction is large enough to matter—including under mechanically constrained configurations where ionic relaxation is limited—and varies with supercell size and dopant chemistry; and (iii) failure tracks covalent-radius mis-

match and elastic frustration rather than donor/acceptor π -electron descriptors alone. We focus on qHP C₆₀ rather than rigid graphene or planar COF hosts because its mixed pentagon-hexagon topology and intrinsically prestrained fullerene cages [13] create competing local and global strain fields in a soft covalent network where bilinear composition-strain terms are expected but have not been audited at periodic scale [3, 4]; substitutional B, N, and P provide a minimal donor, acceptor, and size-mismatch triad (~ 7 – 30 pm covalent-radius excess vs. C; Table VII), not a complete chemical screen. We quantify the omitted bilinear term with matched four-corner audits; $\mathcal{S}$ [Eq. (2)] reports its meV-per-atom value at stated boundary conditions and is *not* proposed as a new physical quantity (Sec. II C). We find that (1) additive strain-dopant models fail for size-mismatched phosphorus, with rigid-lattice upper bounds up to $|\mathcal{S}| = 31.9 \pm 2$ meV/atom at $n=1$ and +3% biaxial strain ($\eta \approx 8\%$ at $n=4$) while ionic relaxation can suppress equilibrium magnitudes by orders of magnitude; (2) linear α does not predict $|\mathcal{S}|$ or its sign across B/N/P; and (3) $|\mathcal{S}|$ tracks frozen local elastic incompatibility more closely than donor/acceptor identity or π -DOS proxies alone (Secs. III and IV). The primary discovery is *physical*: under constrained load on a soft covalent network, stored elastic incompatibility generates a bilinear strain-composition interaction that can break additive high-throughput workflows—not a new descriptor or thermodynamic order parameter. A general rule on the audited triad: non-additivity scales with frozen local elastic incompatibility rather than dopant identity itself (Sec. IV B). On qHP C₆₀, the matched four-corner audit is a protocol that is *likely* transferable only after repeating the same (ϵ, δ, n) grid on each host; we do not claim materials-general laws from a single network.

II. COMPUTATIONAL METHODS

A. Electronic structure method

All total-energy calculations use CP2K [14] with Goedecker-Teter-Hutter (GTH) pseudopotentials, the DZVP-MOLOPT-SR-GTH atom-centered basis, and the Gaussian and plane waves (GPW) scheme.

a. Exchange-correlation and dispersion. We use PBE [15] with Grimme DFT-D3 (Becke-Johnson damping) [16] for the production dataset. Hybrids or meta-GGAs (SCAN, r²SCAN) can shift absolute totals by tens of meV per atom in carbon networks; production rankings are therefore framed as *within-PBE+D3* comparisons of $\mathcal{S}$ among B, N, and P under matched load. The four-corner construction [Sec. II C] subtracts totals computed with identical XC, basis, dispersion, and numerical settings, so systematic shifts largely cancel when *ranking* dopants at fixed protocol—but cancellation is not assumed without audit: on the tetramer P grid, Grimme-D3 non-additivity contributes only $\mathcal{S}_{\text{vdW}} \approx$

+0.33 meV/atom (Table I), far below $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}|$ (Table V). Representative SCAN and r²SCAN four-corner spot-checks on periodic P at $n=1$, +3% are targeted sensitivity tests (Table I; Limitations); no meta-GGA $\mathcal{S}$ values appear in the main text without converged outputs.

b. Plane-wave mesh and Brillouin-zone sampling. Plane-wave cutoffs are 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (REL_CUTOFF 50 Ry; QS_EPS_DEFAULT $10^{-10} E_h$). Comparisons at fixed n always use the same cutoff. Production supercells contain 60–480 atoms with in-plane lattice constants up to ~ 44 Å at $n=8$; inputs use Γ -only sampling. Because $\mathcal{S}$ is a second-order difference on the *same* supercell, uniform k -point or cutoff shifts that co-drifts all four corners largely cancel in the four-corner estimator; sensitivity must be audited on $\mathcal{S}(\text{cutoff})$ and $\mathcal{S}(k)$, not on bare totals alone. A partial cutoff audit at $6 \times \text{C}_{60}$ N and +3% shifts one corner by only ~ 0.057 meV/atom between 350 and 400 Ry, implying $|\Delta \mathcal{S}| < 0.2$ meV/atom when only the mesh drifts (Table I). Dense- k ($2 \times 2 \times 1$) four-corner benchmarks on $n=1$ B/N/P at +3% are pending (Table I); until they converge, the smallest reported $|\mathcal{S}|$ values (2.3–3.0 meV/atom for N and B at $n=1$) are interpreted with explicit Γ -only caution.

c. SCF convergence and four-corner precision. Unless noted otherwise, the SCF threshold is EPS_SCF = $10^{-6} E_h$ in inner and outer orbital-transformation loops (MAX_SCF 300; OUTER_SCF MAX 30). Charged-state workflows in the Supplemental Material tighten this to $10^{-7} E_h$; selected stiff tetramer corners may use $10^{-5} E_h$ when outer SCF cycling persists at 10^{-6} (Table V). Because $\mathcal{S}$ is built from four independently converged totals [Eq. (2)], uncertainty is propagated as $\sigma_{\mathcal{S}} \approx 2\sigma_e$ for uncorrelated per-atom corner errors σ_e (Table I). We use $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom as the conservative reporting band on the $n \leq 4$ production grid.

d. Spin, charge, and mechanical boundary conditions. Substitutional periodic cells are charge-neutral closed-shell Kohn-Sham calculations (no spin polarization in the production $\mathcal{S}$ grid); Hirshfeld strain-path audits on $n=1$ B/N/P use spin-polarized KS only where odd electron counts require it (Sec. II D 3). We do not apply compensating background charge or charged supercells in the $\mathcal{S}$ audit: doping is modeled as neutral substitution, and ΔE_{sub} is not a formation enthalpy (Sec. II B).

Biaxial strain scales lattice vectors at fixed fractional coordinates (rigid-strain protocol). This is a deliberate mechanical *upper bound* relevant when ionic relaxation is kinetically restricted—epitaxial clamping on a rigid substrate, strong interfacial adhesion, or ultrafast biaxial loading—rather than an equilibrium free-energy prediction under fully relaxed internal coordinates (Sec. II D; Discussion, constrained-loading paragraph).

B. Strain and doping models

Biaxial strain is defined as $\epsilon = (a - a_0)/a_0 \times 100\%$, where a (a_0) is the strained (pristine equilibrium) in-plane lattice constant; dopant species $\delta \in \{\text{B, N, P}\}$. The strain–dopant coupling energy $\mathcal{S}$ is defined in Sec. II C [Eqs. (2)–(3)]. π -projected densities of states are written π -DOS.

Tetramer models ($4 \times \text{C}_{60}$) substitute B, N, or P at nominal ~ 5 at.% on a fixed pseudorandom carbon-site map; configurations are not thermodynamically optimized. Periodic supercells place one substituent per C_{60} (~ 1.67 at.%) on a fixed production site map shared across $n = 1$ –8. Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain) is obtained from rigid PBE+D3 tetramer grids; positive α means tensile strain increases the total energy. The substitution energy difference $\Delta E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is not a thermodynamic formation enthalpy (no elemental chemical potentials). Reported α and $\mathcal{S}$ under the rigid-strain protocol (below) are upper bounds on equilibrium magnitudes when ionic relaxation is allowed (Sec. II D).

a. Rigid vs. relaxed mechanical protocols. *Rigid strain* scales in-plane lattice vectors at fixed fractional atomic coordinates: ionic positions do not relax with ϵ . This models mechanically clamped or ultrafast-loaded configurations and yields an upper bound on $|\mathcal{S}|$ when internal stress cannot relax. *Relaxed strain* allows fixed-cell BFGS geometry optimization at each corner before the coupling energy is evaluated on matched coordinates (Table V; Fig. 1(e)); ionic relaxation can reverse or nearly extinguish rigid $\mathcal{S}$ (Sec. II D 2). Force convergence targets are 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr (~ 0.023 and ~ 0.015 eV/Å) at $\epsilon=0$ and +3%, respectively, with matched displacement thresholds (Sec. II D 2); these are tighter than typical 0.02 eV/Å loose-scan defaults because relaxation can drive $|\mathcal{S}|$ toward zero and must be resolved at sub-meV/atom scale.

C. Strain–dopant coupling energy

The physical hypothesis is the bilinear cross term $c\epsilon x$ in Eq. (1) (Introduction): under constrained load, $c \neq 0$ and can be large enough to invalidate additive screening. $\mathcal{S}$ is the symmetric four-corner finite-difference *estimator* of that term at +3% load—not a postulated order parameter (descriptor comparison: Supplemental Material, Table II; thermodynamic caveats: Sec. IV D). Among relaxed free-energy differences ΔG , enthalpic formation energies with chemical potentials, or pairwise binding increments, Eq. (2) is the unique four-corner construction that removes independent strain-only and doping-only contributions in one step.

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (2)$$

with $\Delta E(\epsilon, \delta) = [E(\epsilon, \delta) - E(0, \text{pristine})]/N_{\text{atoms}}$ and $\delta=0$ denoting the pristine reference at the same supercell size $n \times \text{C}_{60}$. Equivalently,

$$\mathcal{S}(\epsilon, \delta) = \frac{E(\epsilon, \delta) - E(\epsilon, 0) - E(0, \delta) + E(0, 0)}{N_{\text{atoms}}}, \quad (3)$$

which depends only on the four corner totals (reference independence). In the continuum limit of small strain and composition increments,

$$\mathcal{S}(\epsilon, \delta) \approx \frac{\partial^2 E}{\partial \epsilon \partial \delta} \Delta \epsilon \Delta \delta + \mathcal{O}(\Delta \epsilon^2, \Delta \delta^2, \Delta \epsilon^2 \delta, \epsilon \delta^2), \quad (4)$$

so $\mathcal{S}$ reports the discrete mixed second derivative of total energy on the audited (ϵ, δ) grid. Only a four-state construction removes linear strain, linear doping, and shared reference offsets in one symmetric estimator; at finite +3% load and discrete B/N/P labels, $\mathcal{S}$ quantifies the nonlinear cross term that additive models drop.

We report $\mathcal{S}$ in meV per atom; $\mathcal{S} < 0$ (> 0) means the coupled corner is more (less) stable than the additive estimate. Strain ϵ and dopant label δ are not interchangeable control axes.

If each corner carries uncorrelated per-atom uncertainty σ_e , error propagation gives $\sigma_{\mathcal{S}} = 2\sigma_e$. Table I lists demonstrated and pending contributions to the reporting band; we use $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom as a conservative protocol floor on the $n \leq 4$ grid when interpreting the smallest periodic points (2.3–3.0 meV/atom for N and B at $n=1$). Periodic $\mathcal{S}$ values in the main text are from the +3% size-scaling grid (Table II).

To quantify additive failure magnitude we report

$$\eta \equiv \frac{|\mathcal{S}|}{|\Delta E_{\text{coupled}}|}, \quad (5)$$

where $\Delta E_{\text{coupled}} = (E(\epsilon, \delta) - E(0, 0))/N_{\text{atoms}}$. At $n=4$ and +3% strain, $\eta \approx 8\%$ for P ($|\mathcal{S}| = 23.7$ meV/atom vs. $|\Delta E_{\text{coupled}}| \approx 283$ meV/atom), whereas B and N remain below 1%.

a. Link to mechanical protocols (Secs. II B and II D 2). All $\mathcal{S}$ values quoted for screening comparisons use the rigid-strain protocol unless Table V or Fig. 1(e) states otherwise. Relaxed-strain $\mathcal{S}$ is evaluated on ionically optimized coordinates at fixed biaxial cell shape; it can reverse sign relative to the rigid upper bound because internal cage stress is partially released before the four-corner difference is taken.

D. Validation benchmarks

Four protocol checks govern interpretation: (i) **four-corner numerical hygiene**—Table I ($\sigma_{\mathcal{S}}$ propagation, partial cutoff and vdW audits); (ii) **ionic relaxation**—tetramer P four-corner fixed-cell BFGS and periodic $n=1$ P geometry optimization (Table V; Fig. 1(e));

TABLE I. Error budget for four-corner $\mathcal{S}$ [Eq. (3)] on the production grid. $\sigma_S \approx 2\sigma_e$ assumes uncorrelated per-atom corner errors σ_e ; identical XC, k -point, and cell settings across all four corners induce partial cancellation, so realized noise is often smaller. Rows marked “pending” are targeted sensitivity tests (Limitations); no meta-GGA or dense- k numbers appear without converged outputs. The reporting band (± 2 meV/atom) is a conservative protocol floor, not a quadrature sum of independent rows.

Source	$ \Delta\mathcal{S} $ or σ_S (meV/atom)
SCF convergence (EPS.SCF = 10^{-6})	$\lesssim 2$ ($\sigma_S \approx 2\sigma_e$)
Plane-wave cutoff (350 vs 400 Ry)	< 0.2
D3 dispersion cross term	$ \mathcal{S}_{\text{vdW}} \approx 0.33$
Ionic relaxation (rigid vs. relaxed P)	$\gg 2$ (orders of magnitude)
k -point (Γ vs $2 \times 2 \times 1$)	—
XC (PBE+D3 vs SCAN / r ² SCAN)	—
<i>Adopted reporting band ($n \leq 4$): ± 2 meV/atom.</i>	

(iii) **matched-functional rigid vs. relaxed** coordinates on the tetramer P grid (Table V); and (iv) **alternate dopant placement** on independent site maps (Table II).

Tetramer corners use the BFGS thresholds quoted above (Sec. IID 2). Periodic ionic benchmarks include $n=1$ P on the production map (largest rigid $|\mathcal{S}|$); tetramer Table VII supplies the complementary ~ 5 at.% relaxation geometry. Periodic $n=4$ P fixed-cell relaxation, B/N periodic relaxations, and dense- k / meta-GGA four-corner spot-checks remain targeted backlogs (Table I; Limitations). Size scaling extends to $n=8$ (480 atoms, in-plane lattice ~ 44 Å). With one substituent per C_{60} and vdW-separated cages, electrostatic dilution and lateral periodicity make spurious cell-cell coupling subdominant to cage-local misfit stress on the audited maps, but explicit two-body periodic-image convergence tests on $|\mathcal{S}|$ beyond this grid were not performed.

The task-group inventory in Table V (superseded exploratory grids in Table IV) lists N_{conv} and main-text linkage; ionic-relaxation energies appear in Table VII; the alternate-placement grid in Table II spans the full B/N/P strain path and six modern PBE+D3 pristine references. Three validation benchmarks—ionic-relaxation tetramer $\mathcal{S}_P$ (Table VII), alternate dopant placement (Table II), and a cutoff benchmark at $6 \times \text{C}_{60}$ N @ +3% (Table V)—govern interpretation of size-scaling trends (Secs. III A and IV).

Sec. IIIC reports E_{sub} only when tied to the verified PBE+D3 grids above (no chemical-potential reference; not a formation enthalpy).

1. Alternate-placement benchmark

An independent dopant configuration at the same nominal ~ 5 at.% loading repeats the B/N/P strain grid under rigid PBE+D3 (Table VI). Main-text quantitative α and $\mathcal{S}_{+3\%}^{\text{tet}}$ use the reference-placement map

in Fig. 3(a,b); the alternate seed 137 grid is archived only. At periodic $n=4$, two independent site maps repeat the four-corner Eq. (2) audit; the $|\mathcal{S}|$ ranking at +3% matches the production map in Table II.

2. Ionic-relaxation benchmark

Fixed-cell BFGS geometry optimization is performed on the tetramer pristine/ P corners at $\epsilon=0$ and +3% (force thresholds 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr, 0.023 and 0.015 eV/Å; displacement thresholds 3.0×10^{-3} and 1.5×10^{-3} Å). These thresholds are tighter than typical 0.02 eV/Å defaults because relaxation can drive $|\mathcal{S}|$ toward zero and must be resolved at sub-meV/atom scale. A matched-functional rigid PBE+D3 control at the relaxed coordinates quantifies how $\mathcal{S}$ changes upon relaxation (Table V). The same BFGS thresholds are used for the periodic $n=1$ P four-corner relaxation reported in Fig. 1(c).

3. Hirshfeld population along strain

Hirshfeld atomic charges are computed along $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ on periodic $n=1$ B/N/P cells (spin-polarized KS for odd electron counts). They complement geometric $\bar{d}$ in Table VII.

III. RESULTS

We report total-energy audits on matched periodic supercells and on the reference tetramer map under mechanically constrained joint load (Sec. IIC). Observations below are stated without mechanistic interpretation; Sec. IV addresses origins, screening limits, and literature context.

A. Non-additive strain–doping coupling

Figure 1 summarizes the central test: separate strain-only and substitution-only DFT increments are summed additively, compared with the explicitly coupled four-corner total, and the signed gap $\mathcal{S}$ [Eq. (3)] quantifies when that additive workflow mis-ranks stability (panel f). Panel (c) compares, at $n=4$ and +3% biaxial strain, the additive prediction $\Delta E_{\text{strain}} + \Delta E_{\text{dop}}$ with the explicitly coupled four-corner increment; their signed difference is $\mathcal{S}$. The coupled bars exceed the additive bars for B, N, and P; P shows the largest gap ($\mathcal{S} = -23.7 \pm 2$ meV/atom; $\eta \approx 8\%$). Panels (a,b) map $\mathcal{S}(n, \delta)$ on the production grid and size-scaling trajectories with provisional $\mathcal{S}_{\infty} + A/n$ guides. Across fifteen periodic +3% points [Fig. 1(d)], $|\mathcal{S}|$ spans 2.2–31.9 meV/atom (median 7.8 meV/atom; quartiles marked); the P value at

TABLE II. Periodic coupling-energy audit at +3% biaxial strain (PBE+D3 rigid protocol, reference placement; core $n \leq 4$ at 400 Ry). $\mathcal{S}$ in meV/atom from Eq. (2); conservative protocol uncertainty ± 2 meV/atom for $n \leq 4$ (Sec. IIC). Rows with $n \geq 6$ (350 Ry) are in Supplemental Material Table S1.

n (C_{60} units)	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
1	B	+3	-8.38
1	N	+3	+2.30
1	P	+3	-31.91
2	B	+3	-2.18
2	N	+3	+6.85
2	P	+3	-28.29
4	B	+3	-6.23
4	N	+3	+2.95
4	P	+3	-23.71

TABLE III. DFT calculation groups supporting the main-text claims (reproducibility inventory; Supplemental Material only). Superseded exploratory grids appear in Table IV.

Calculation group	System	Count	Protocol
Archived baseline (legacy PBE)	$4 \times C_{60}$	24	Rigid strain; not used for main-text claims
π -DOS and gaps	$4 \times C_{60}$	12	PBE+D3
IPR (polaron dimer)	Dimer	4	Pristine/B; $\epsilon=0, +3\%$
Hopping J	Dimer	4	PBE+D3
Charged polarons ^a	$4 \times C_{60}$	20	UKS; geometry opt. + vertical SP
Periodic size scaling	$n \times C_{60}$	41	PBE+D3; 400/350 Ry
Cutoff benchmark	$6 \times C_{60}$ N	1	+3%; 400 Ry single-point
Ionic relaxation	$4 \times C_{60}$	4	Fixed-cell GEO; P corners
Ionic relaxation	$1 \times C_{60}$ P	4	Fixed-cell GEO; periodic
Alternate placement	$4 \times C_{60}$	24	Rigid PBE+D3
Periodic placement	$4 \times C_{60}$	16	$n=4$; two independent maps
Matched-functional rigid	$4 \times C_{60}$	4	PBE+D3 at relaxed coordinates
Reference placement	$4 \times C_{60}$	24	Rigid PBE+D3
Hirshfeld population	$1 \times C_{60}$ B/N/P	18	Spin-polarized KS

^aSix relaxed tetramer geometries provide charged-state initial structures.

$n=1$ (31.9 ± 2 meV/atom) is an extremum, not a typical audit magnitude. Internal protocol audits bound $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom on the $n \leq 4$ grid (Sec. IIC).

a. Operational significance thresholds (this work). We flag additive failure when two *adopted* criteria are met on the audited grid: (i) $|\mathcal{S}| > \sim 2$ meV/atom, above the conservative four-corner protocol band, and (ii) $\eta = |\mathcal{S}|/|\Delta E_{\text{coupled}}| > \sim 1\%$ on the same supercell. These cut-offs are operational, not universal; P meets both at $n=4$ ($\eta \approx 8\%$), whereas B and N do not on the production map. At $n=4$ (Table II), $\mathcal{S} = -6.2 \pm 2$ meV/atom (B), $+3.0 \pm 2$ meV/atom (N), and -23.7 ± 2 meV/atom (P), where $\pm$ denotes a conservative protocol upper bound. Within $n \leq 4$, $|\mathcal{S}|$ is not monotonic in n ; we report protocol values without extrapolating $\mathcal{S}_{\infty}$. Rows with $n \geq 6$ (350 Ry) appear in Supplemental Material Table S1. Negative $\mathcal{S}$ (B, P at $n=4$) means the coupled corner lies below the additive estimate under the stated boundary conditions; positive $\mathcal{S}$ (N) means the coupled corner lies above it. Panel (e) shows one illustrative rigid-vs.-relaxed pair for $n=1$ P at +3% ($\mathcal{S}_{\text{rig}} \approx +31.9$ meV/atom vs. near-zero $\mathcal{S}$ after ionic relaxation; Table V); the near disappearance of $|\mathcal{S}|$ upon relaxation indicates that the

audited coupling originates predominantly from frozen elastic incompatibility rather than from an intrinsic electronic interaction term. At $n=1$ and +3%, $|\mathcal{S}|$ increases monotonically with $|\Delta r_{\text{cov}}|$ for B and P (7 and 30 pm $\rightarrow$ 8.4 and 31.9 meV/atom) but N (6 pm) breaks a simple radius-only scaling (2.3 meV/atom), consistent with a secondary electronic channel (Fig. 4(d); Sec. IV B). Table V lists the matched-functional tetramer benchmark ($\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ vs. $\mathcal{S}_{\text{relaxed}} = -2.28$ meV/atom; sign not preserved).

B. Electronic structure under strain and doping

Figure 2(a) shows pristine and B/N/P-substituted tetramer cages at $\epsilon=0$ with highlighted substituent sites; covalent-radius excess Δr_{cov} is listed in Table VII. Panels (b)–(d) report one-parameter slices used in sequential screens: the N π -DOS waterfall shifts with biaxial strain; $E_g(\epsilon)$ paths place B ($E_g \approx 0.03$ eV), N ($E_g \approx -0.14$ eV), and P ($E_g \approx 0.05$ eV) at $\epsilon=0$ on distinct trajectories while pristine remains open; panel (d) gives the pristine mechanical $\Delta E(\epsilon)$ path on the reference strain grid. Sup-

TABLE IV. Exploratory DFT grids superseded by the production protocol (reproducibility archive only).

Task group	System	N_{conv}	Disposition
Early strain probe	$2 \times \text{C}_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2 \times \text{C}_{60}$	24	Superseded
Exploratory design grid	$4 \times \text{C}_{60}$	20	Archive only

TABLE V. Ionic-relaxation benchmark for the tetramer P -substitution four-corner grid (fixed-cell geometry optimization: pristine and P at $\epsilon=0$ and $+3\%$, same biaxial cell). Relaxed corners and the matched-functional ratio use PBE+D3 ($\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom; $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$; sign not preserved). $\mathcal{S}$ follows Eq. (2) with $N=240$ atoms.

Quantity	Rigid (PBE+D3 matched)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_{\text{h}})$	—	−1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_{\text{h}})$	—	−1363.9550
$E(P, \epsilon=0) (E_{\text{h}})$	—	−1374.4415
$E(P, \epsilon=3\%) (E_{\text{h}})$	—	−1374.4615
$\mathcal{S}$ (meV/atom)	+1.19	−2.28
Sign of $\mathcal{S}$ vs. rigid	+	− (not preserved)

plemental Material gives full π -DOS stacks and periodic VBM/CBM isosurfaces (N size series; P $n=4$): frontier density remains cage-localized for $n = 2-6$.

C. Mechanical strain sensitivity

Pristine tetramers yield $|\alpha_{\text{pris}}| \approx 1$ meV/% on the six-point reference grid. Least-squares slopes for B, N, and P are $+58.7$, -297.5 , and $+456.0$ meV/% [Fig. 3(d); horizontal bars give linear-fit standard errors on that grid].

Table VII reports $\bar{d}$, $\sigma(\bar{d})$, and Δr_{cov} at representative strains. At $+3\%$, P shows $\Delta \bar{d} \approx 3 \times 10^{-5}$ Å versus $\sim 8.5 \times 10^{-3}$ Å for B; on the response plane in Fig. 4, P carries the largest $|\mathcal{S}|$ bubble and is the only dopant with $|\Delta r_{\text{cov}}| \gtrsim 20$ pm.

Figure 3 traces microscopic strain paths on the reference map. Hirshfeld $q(\epsilon)$ and $\bar{d}(\epsilon)$ [panels (a,b)] vary with dopant and strain. Local disorder $\sigma(\bar{d})(\epsilon)$ [panel (c)] broadens for B and N under tension while remaining near the $\epsilon=0$ baseline for P. Panel (d) maps reference-map $|\alpha|$ against periodic $|\mathcal{S}|_{n=4}$ (marker area $\propto |\mathcal{S}|_{n=1}$): N lies in the high- $|\alpha|$ /low- $|\mathcal{S}|_{n=4}$ quadrant and P in the high- $|\mathcal{S}|$ region—a qualitative rank inversion among three dopants ($n=3$), not a fitted correlation law.

Among audited substituents, P has the largest covalent-radius mismatch ($\Delta r_{\text{cov}} = +30$ pm; Table VII) and the largest audited $|\mathcal{S}|$ under the rigid loading protocol; Sec. IV B decomposes that pattern.

IV. DISCUSSION

A. Main finding

The present results indicate that additive strain-dopant energetic screening fails under mechanically constrained joint load on qHP C_{60} : totals on the $E(\epsilon, \delta)$ landscape need not equal the sum of separate strain-only and substitution-only increments, and the bilinear coefficient in Eq. (1) is nonzero in the four-corner sense [Eq. (4)]. The calculations suggest that, for phosphorus at $+3\%$ biaxial load, the omitted term is large enough to change high-throughput stability decisions ($|\mathcal{S}| = 31.9 \pm 2$ meV/atom at $n=1$; 23.7 meV/atom at $n=4$; Table II), while linear α does not rank $|\mathcal{S}|$ or its sign [Fig. 3(d)]. $\mathcal{S}$ is the finite-difference estimator of $\partial^2 E / \partial \epsilon \partial \delta$ on the audited grid—a diagnostic of additive failure, not a new physical quantity. Ionic relaxation nearly extinguishes $|\mathcal{S}|$ (Table V; Fig. 1(e)), so the near disappearance of the cross term upon stress release indicates that the audited coupling originates predominantly from frozen elastic incompatibility rather than from an intrinsic electronic interaction term; rigid-strain $\mathcal{S}$ therefore bounds constrained or kinetically limited load (Sec. IV E 0 a).

Significant additive failure is flagged operationally by the thresholds adopted in Sec. III A: $|\mathcal{S}| > \sim 2$ meV/atom (above protocol noise) and $\eta > \sim 1\%$ (non-negligible relative to the coupled corner increment), read alongside the $|\mathcal{S}|$ distribution rather than a single extremum. These thresholds are practical screening criteria rather than thermodynamic boundaries.

a. When additive screening should fail. On general grounds, coupled audits are expected to matter when (i) local substitutional strain is large and cannot relax (size mismatch, rigid substrate, or short loading times);

TABLE VI. Archived alternate-placement tetramer grid (PBE+D3 rigid; seed 137). Not used for main-text figures or quantitative claims; retained for reproducibility of the placement-sensitivity audit. Reference-map α and $\mathcal{S}_{+3\%}^{\text{tet}}$ appear in Fig. 3(a,b). Periodic $\mathcal{S}(n)$ is in Table II.

Dopant	Placement	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)
B	alternate	+21.4	-2.6
N	alternate	+45.4	+12.6
P	alternate	+989.6	+41.4

f

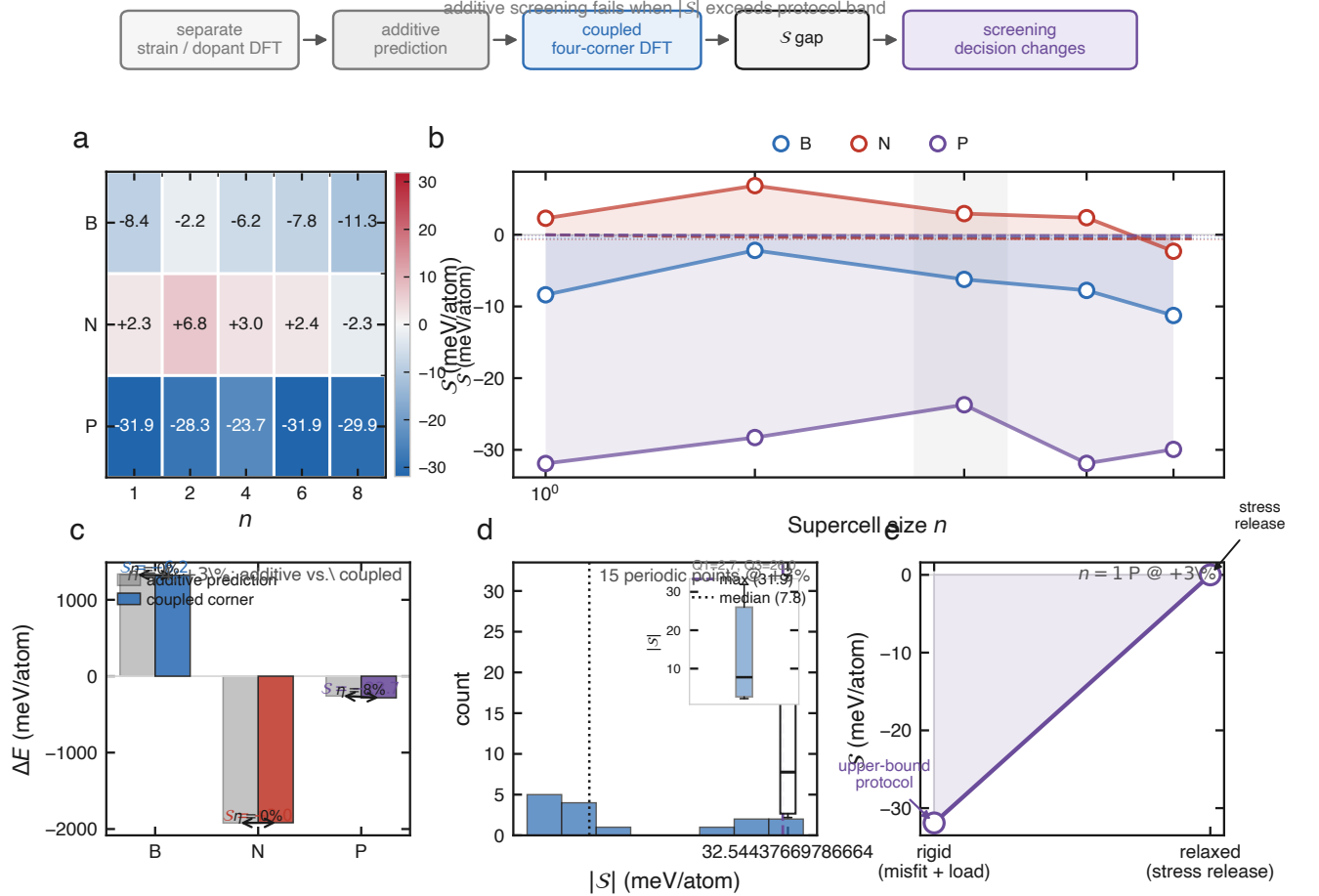


FIG. 1. When additive strain-dopant screening fails (graphical summary and audit). (f) Workflow: separate strain-only and substitution-only DFT increments are summed additively, compared with the explicitly coupled four-corner total; the gap $\mathcal{S}$ [Eq. (3)] flags when stability rankings change at meV-per-atom margins. (c) At $n=4$ and $+3\%$: additive prediction vs. coupled four-corner ΔE ; arrows label $\mathcal{S}$ and fractional error η [Eq. (5)]. (a,b) $\mathcal{S}(n, \delta)$ heatmap and size-scaling trajectories with provisional $\mathcal{S}_{\infty} + A/n$ guides. (d) Distribution of $|S|$ over fifteen periodic $+3\%$ points (histogram; box: median and quartiles; max 31.9 meV/atom marked). (e) Illustrative rigid vs. relaxed path for $n=1$ P only ($+3\%$): stored elastic work under constrained load vs. stress release after ionic relaxation.

(ii) bonding is short-range and elastically soft, so defect fields overlap under external load; and (iii) hosts exhibit elastic anisotropy or intrinsic pre-strain, as in buckled covalent molecular networks [13, 17]. qHP C_{60} satisfies these conditions; stiffer two-dimensional hosts may require separate benchmarks.

B. Mechanism

For size-mismatched P, the calculations suggest a single dominant causal chain (Fig. 4): large covalent-radius mismatch ($\Delta r_{\text{cov}} = +30$ pm) $\rightarrow$ frozen local misfit stress in a pre-buckled cage $\rightarrow$ cooperative overlap with external biaxial tension $\rightarrow$ bilinear elastic work $\rightarrow$ largest $|S|$. *Primary* control is geometric misfit strain and elastic

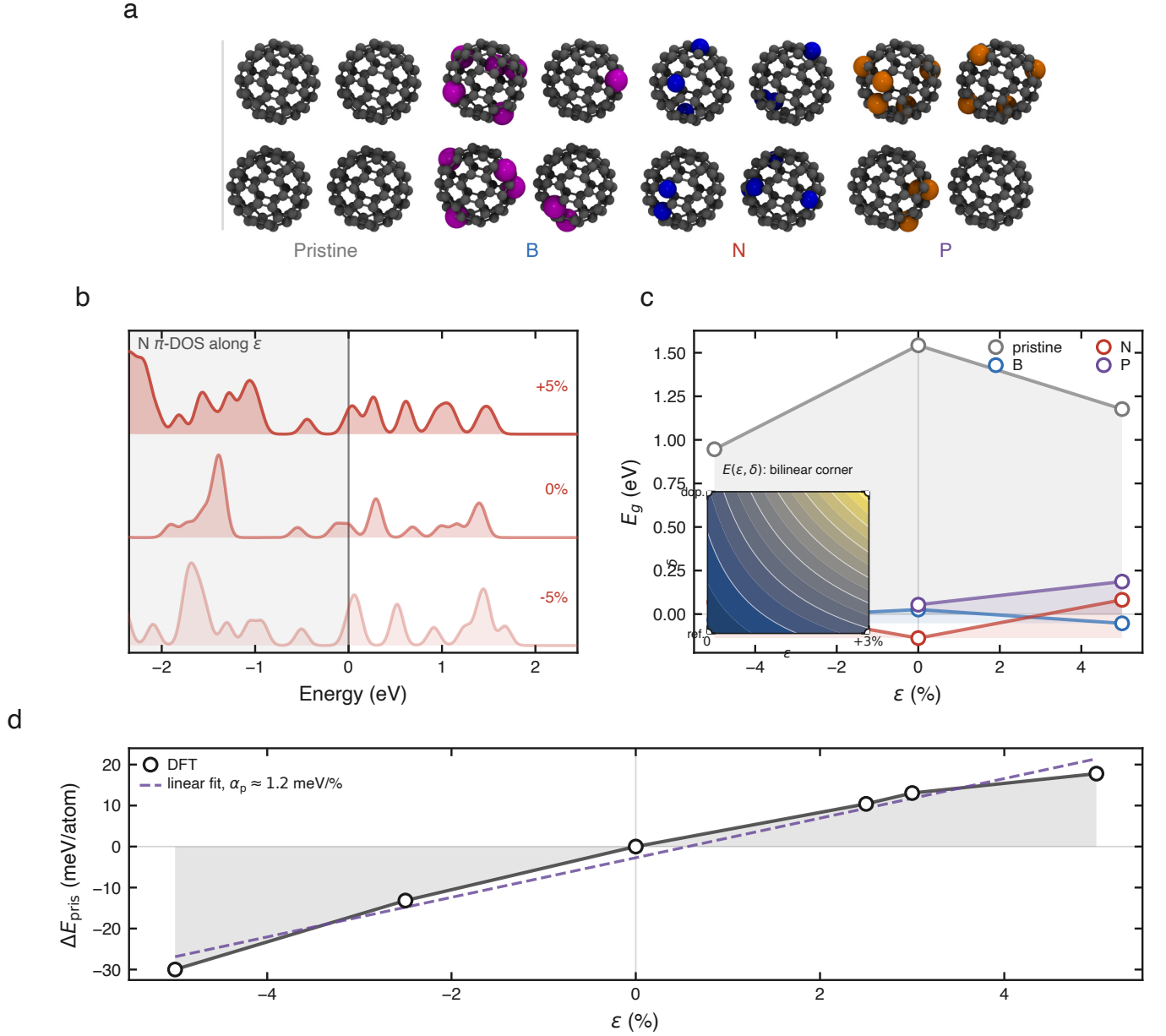


FIG. 2. Reference tetramer structure and one-parameter strain slices (PBE+D3). (a) qHP C₆₀ cages at $\epsilon=0$: pristine and B/N/P substitution sites (Δr_{cov} in Table VII). (b)–(d) Sequential-screen baselines on the reference map: N π -DOS waterfall; $E_g(\epsilon)$ with schematic $E(\epsilon, \delta)$ bilinear corner (inset); pristine mechanical $\Delta E(\epsilon)$. Joint-load four-corner audits of the omitted cross term appear in Fig. 1.

frustration; *secondary* is electronic redistribution (Hirshfeld Δq on P is moderate, $\sim 11\%$ fractional $|\Delta q/q|$, and does not exceed N's $\sim 18\%$); *tertiary* is π -electron / π -DOS morphology. Electronic structure responds to the coupling rather than controlling it: π -DOS tracks donor–acceptor character and α sign but does not, by itself, rank periodic $|\mathcal{S}|$ at $n=4$ [Fig. 2(b,c); Fig. 3(d)].

a. General rule. On the audited B/N/P triad, non-additivity scales with frozen local elastic incompatibility rather than dopant identity itself: $|\mathcal{S}|$ rises with $|\Delta r_{\text{cov}}|$ for B and P at $n=1$ (7 and 30 pm $\rightarrow$ 8.4 and

31.9 meV/atom), while N breaks a radius-only trend (6 pm, 2.3 meV/atom), signalling a secondary electronic channel. A qualitative design coordinate is $|\Delta r_{\text{cov}}| \gtrsim 20$ pm for the geometric-nonlinear regime (P); $|\Delta r_{\text{cov}}| \lesssim 15$ pm leaves $|\mathcal{S}|$ at few-meV/atom but still permits α – $\mathcal{S}$ sign conflicts (B, N).

Substitutional B, N, or P embed a localized internal strain field into an already distorted qHP cage baseline [13]. External biaxial strain is a uniform macroscopic load; sequential models assume that responses superpose linearly, but $\mathcal{S}$ shows that second-order defect–

TABLE VII. Local dopant–cage structure from archived *tetramer* XYZ ($\sim 5\%$ nominal doping; qualitative proxy for mismatch trends). Mean nearest-neighbor C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm. Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon = 0)$ tracks the uniform nearest-neighbor stretch component. Distances are DFT-derived from archived tetramer XYZ; no experimental bond errors are quoted—comparison is internal across dopants and with qHP literature on intrinsic cage strain [10, 13].

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon = +3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

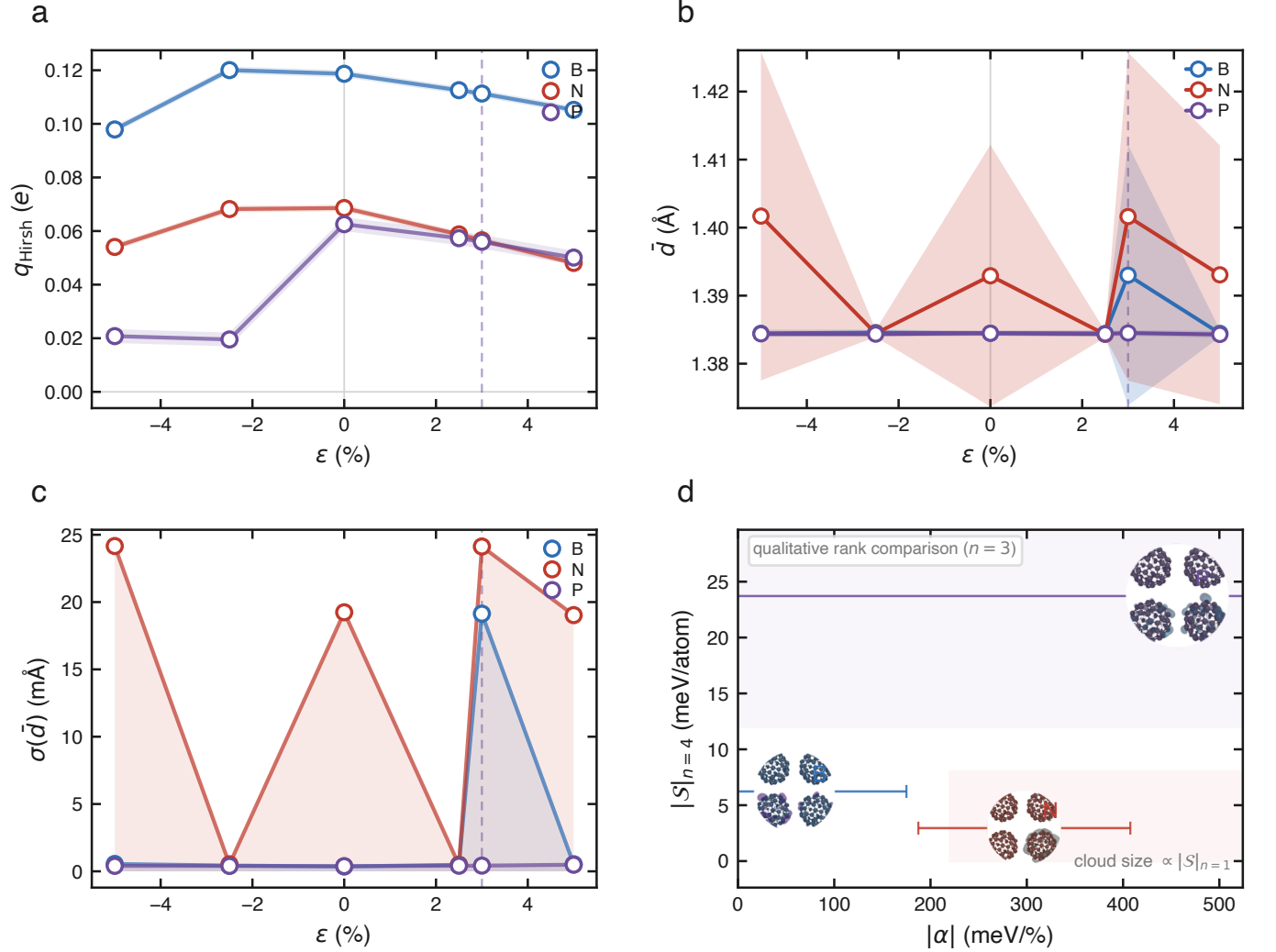


FIG. 3. Microscopic strain paths and α – $|S|$ rank comparison ($n=3$ dopants). (a) Hirshfeld $q(\epsilon)$ for B/N/P (six-point paths; dotted guide at +3%). (b) Nearest-neighbor $\bar{d}(\epsilon)$ with $\pm\sigma$ bands. (c) Local disorder path $\sigma(\bar{d})(\epsilon)$. (d) Reference-map $|\alpha|$ vs. periodic $|S|_{n=4}$ (marker area $\propto |S|_{n=1}$; horizontal bars: linear-fit standard errors on the six-point strain grid). N inverts a naive $|\alpha| \rightarrow |S|$ expectation; with only three dopants this panel reports qualitative ranking, not Pearson, R^2 , or Spearman statistics.

elastic work can exceed linear $\alpha \times \epsilon$ scaling on this soft, pre-strained host. N introduces excess π electrons (negative α); B creates electron deficiency (positive α); the α – $|S|$ rank mismatch is that large $|\alpha|$ does not imply large periodic $|S|$ (N vs. P at $n=4$). B shows the largest mean bond stretch at +3% ($\Delta\bar{d}$) yet modest $|S|$, whereas

P shows the largest $|S|$ with the smallest $\Delta\bar{d}$ (Table VII). Tetramer α and periodic S probe different concentrations and boundary conditions and need not share sign.

Ionic relaxation releases misfit stress cooperatively with external tension, collapsing rigid $|S|$ toward equilibrium—a stress-release mechanism, not a contra-

diction of nonlinear coupling (Fig. 1(e)). For B and N at $|\Delta r_{\text{cov}}| \lesssim 10$ pm, $|\mathcal{S}|$ is set more by electronic nonlinearity than by large-amplitude cage reshaping. Grimme D3 dispersion at the four tetramer P corners contributes $\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom (Table V), far below $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}|$, so vdW non-additivity does not drive the reported cross terms.

A compact taxonomy on the audited triad: *electronic* donor/acceptor control of α (B, N); *structural* local-distortion coupling to external strain (P); *finite-size* electrostatic dilution (N at large n). At $n=4$ and +3% strain, $|\mathcal{S}| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential additive estimates differ by approximately a factor of four in the load-dependent margin under the same protocol.

C. Relation to literature

The present work differs from previous studies in that it reports a periodic *total-energy* audit of joint biaxial (ϵ, δ) load [Eq. (2)] on qHP C_{60} (Table II), rather than strain-only, doping-only, mobility, or gap-centric mapping [1, 2, 4–7, 9, 10, 12, 13, 18–23]. Prior qHP surveys emphasize elastic moduli and optical gaps [10]; graphendofullerene derivatives tune Curie temperature through strain *or* electrostatic doping in isolation [19]; half-semiconductor engineering targets gaps [7]. The present audit quantifies the omitted additive cross term in total energy when both fields are applied jointly.

We do not claim that nonlinear strain–dopant coupling is unique to qHP C_{60} . This host is experimentally accessible, elastically soft, and intrinsically pre-strained [13], so bilinear composition–strain terms expected in effective Hamiltonians [3] can be audited at periodic scale; stiffer graphene-like or COF hosts [17] may show smaller cross terms and require separate benchmarks. The dopant set is a minimal triad spanning both *electronic* (donor/acceptor) and *geometric* (size mismatch) degrees of freedom: B, N, and P (~ 7 –30 pm covalent-radius excess vs. C; Table VII), not a complete chemical screen. Prior graphullerene work often relaxed ions before comparing totals or sampled strain and substitution on separate grids combined additively [5, 7, 9], masking transient nonlinear coupling under mechanically constrained conditions (Sec. IV E 0 a).

D. Limitations

Main-text α and $\mathcal{S}$ use PBE+D3 under the rigid-strain protocol; ionic relaxation can reverse or nearly extinguish $\mathcal{S}$ (Table V). All totals are 0 K Born–Oppenheimer energies without vibrational or configurational entropy; finite- T rankings could reorder sub-meV competitors but would not restore additivity if bilinear coupling persists at the electronic-structure level. Production periodic cells are charge-neutral closed-shell Kohn–Sham totals;

spin polarization applies only where odd electron counts require it (Sec. II D 3). Geometry optimizations terminate at local BFGS minima; alternate placements (Table II) bound configuration specificity. Core claims use matched 400 Ry for $n \leq 4$; $n \geq 6$ uses 350 Ry; Γ -only sampling, SCAN/ r^2 SCAN spot checks, and dense- k benchmarks remain targeted backlogs (Table I). Table VII geometry is a tetramer proxy; channel assignment rests on $\Delta \vec{d}$, Δr_{cov} , and Hirshfeld Δq rather than Mayer/Bader partitioning. With only three substitutional labels, the α – $|\mathcal{S}|$ map [Fig. 3(d)] is a qualitative rank comparison, not a fitted correlation law. Claims are scoped to qHP C_{60} on fixed site maps.

Several non-physical mimics of a four-corner cross term are excluded by protocol controls: $|\mathcal{S}|$ is not monotonic in n ; all corners are SCF-converged under identical settings; matched cutoffs shift one corner by only ~ 0.057 meV/atom at $6 \times \text{C}_{60}$ N; shared functional and cell shape cancel spurious offsets in Eq. (3), with $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom (Sec. II C).

E. Future work and outlook

The matched $\mathcal{S}$ grid forecasts whether sequential strain-plus-doping screens will mis-rank substitutional stability before expensive relaxations or transport calculations; $|\mathcal{S}| > \sim 2$ meV/atom at +3% flags mandatory coupled re-evaluation. Rigid-strain $\mathcal{S}$ matters when boundary conditions or kinetics prevent full cage relaxation—epitaxial clamping, substrate adhesion, ultrafast or cyclic loading, or nanoscale flexures (Sec. IV E 0 a)—where sequential screens can mis-rank stability by meV/atom margins comparable to the audited distribution (median 7.8 meV/atom across fifteen periodic +3% points; Fig. 1(d)).

Laboratory handles on the stress-field picture include Raman/IR cage-mode shifts that deviate from additive strain-plus-doping predictions, strain-dependent scanning-tunneling spectra when charge redistributes without proportional $\Delta \vec{d}$, and stability trends biased by the sign of $\mathcal{S}$ at $n=4$. Quantitative links require matched sample chemistry and strain calibration.

Targeted follow-up includes Mayer/Bader strain-path partitioning, additional III/V substituents on matched maps, $\mathcal{S}(\text{cutoff})$, $\mathcal{S}(k)$, and $\mathcal{S}(\text{EPS.SCF})$ sensitivity curves, SCAN or r^2 SCAN spot-checks on high- $|\mathcal{S}|$ corners, and periodic $n=4$ P ionic relaxation on the production map.

a. Physical relevance of mechanically constrained loading. Periodic $n=1$ P at +3%: -31.9 meV/atom rigid vs. ~ 0.1 μeV /atom relaxed at matched PBE+D3 (Table V). Rigid-strain $\mathcal{S}$ is therefore not an equilibrium free-energy difference; it bounds omitted coupling when internal stress remains frozen. Fig. 1(e) illustrates stress preservation under rigid load vs. release upon relaxation—supporting, not refuting, the nonlinear coupling picture.

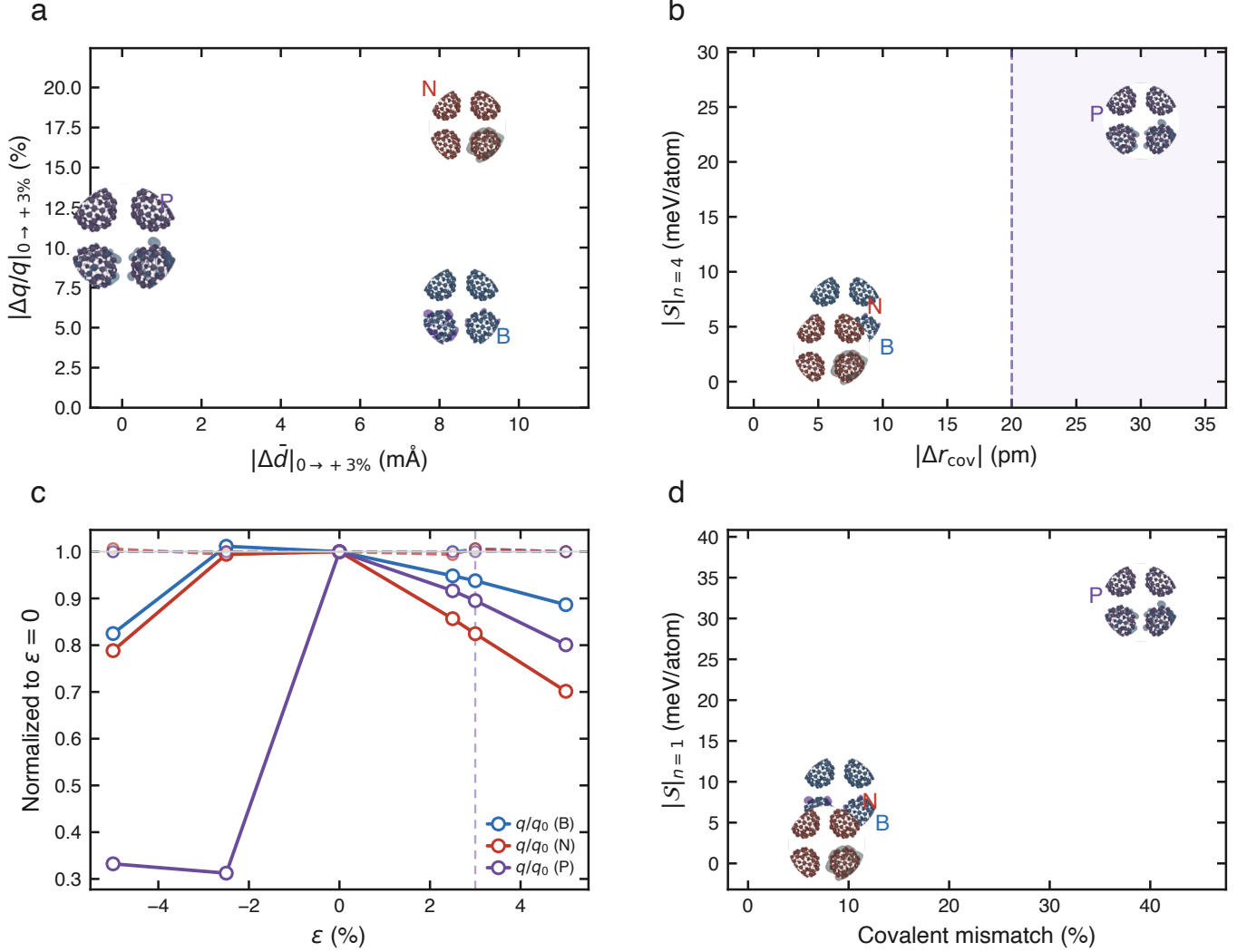


FIG. 4. Mechanism hierarchy for P-dominated coupling (primary: misfit strain $\rightarrow$ frozen internal stress; secondary: charge redistribution; tertiary: π -DOS morphology). (a) Response plane from 0 $\rightarrow$ +3% path endpoints; bubble area $\propto |S|_{n=4}$ (B: large $\Delta \bar{d}$, small $|S|$; N: large $|\Delta q|$, moderate $|S|$). (b) $|\Delta r_{\text{cov}}|$ vs. $|S|_{n=4}$: radius alone does not rank B/N/P. (c) Linked normalized paths $q(\epsilon)/q(0)$ (solid) and $\bar{d}(\epsilon)/\bar{d}(0)$ (dashed). (d) Covalent mismatch (%) vs. $|S|_{n=1}$ (P extremum at largest mismatch).

V. CONCLUSIONS

Nonlinear strain-dopant energetic coupling—not the symbol $\mathcal{S}$ —is the primary finding: additive composition-strain screening fails under constrained load on qHP C₆₀ because stored elastic incompatibility generates a bilinear interaction large enough to reorder stability rankings at meV-per-atom margins ($|S| > \sim 2$ meV/atom and $\eta > \sim 1\%$; Sec. III A). The matched four-corner difference $\mathcal{S}$ quantifies that failure [Eqs. (4)–(3)] as a discrete mixed-derivative estimator, not a new thermodynamic observable. On qHP C₆₀, donor/acceptor dopants produce an α sign split [Fig. 3(d)] that does not reliably predict $|S|$, while periodic $|S|$ reaches 31.9 ± 2 meV/atom for P at $n=1$ at fixed coordinates—an extremum within a broader 2.2–31.9 meV/atom audit distribution (Fig. 1(d)) with

$\eta \approx 8\%$ at $n=4$. Non-additivity scales with frozen local elastic incompatibility rather than dopant identity alone; for P, geometric misfit strain is primary, electronic redistribution secondary, and π -DOS morphology tertiary (Sec. IV B; Fig. 4). Ionic relaxation nearly extinguishes $|S|$, confirming an elastic-work origin under constrained load rather than an intrinsic electronic interaction term (Table V; Fig. 1(e)). Coupled (ϵ, δ) screening is therefore advisable for $|\Delta r_{\text{cov}}| \gtrsim 20$ pm substituents and +3% loads; four-corner audits can pre-screen such cases before full relaxation (Sec. IV E). Generalization beyond qHP C₆₀ is *likely* applicable only after matched four-corner audits on each host; the present study reports the failure condition and audit protocol on one soft covalent molecular network and does not claim universal constants or materials-general laws. Prior strain-doping studies often

masked this coupling by fully relaxing structures before comparing totals or by assembling separate strain-only and doping-only scans additively (Sec. IV C). The theoretical landscape is $E(\epsilon, \delta)$; $\mathcal{S}$ is its quantitative diagnostic under stated boundary conditions, not the primary physical claim.

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DATA AVAILABILITY

The data that support the findings of this study are openly available at [https://github.com/](https://github.com/chenxingqiang/graphullerene-strain-engineering)

[chenxingqiang/graphullerene-strain-engineering](https://github.com/chenxingqiang/graphullerene-strain-engineering), including computational inputs and outputs, tabulated coupling-energy and polaron analyses, and materials to reproduce all figures and tables in the main text. Ionic-relaxation, Hirshfeld, and size-scaling archives referenced in Tables VI–II are included in the public release. A tagged release matching the manuscript revision will be issued upon acceptance.

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